

Week 3: RAG Generation Baseline

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1. Objective

Build an honest end-to-end RAG generation baseline on top of the Week 2 BM25 results. The goal is to measure how well a vanilla pretrained Seq2Seq model answers MS MARCO QA questions when conditioned on the *real* top-k passages returned by BM25 (rather than the toy 3-passage corpus used in the prototype).

2. Pipeline

```
query
  ↓
BM25 retrieval (Week 2 run.tsv)
  ↓
top-3 passages
  ↓
prompt: "question: <q> context: <p1> <p2> ..."
  ↓
t5-small (no fine-tuning)
  ↓
answer
```

3. Dataset

- Queries: MS MARCO Passage Ranking dev/small (intersected with the QA v2.1 validation split so that human-written reference answers are available).
- Reference answers: MS MARCO QA v2.1 validation, answers field (with "No Answer Present." filtered out).
- Number of evaluated queries: **200**.

4. Model

- **t5-small** loaded from HuggingFace, used in pretrained form (no fine-tuning).
- Max input tokens: 512; max new tokens: 64.

5. Evaluation Metrics

All four metrics use **best-of-N references**: when MS MARCO provides multiple human-written answers for a query, each prediction is scored against every reference and the highest score is kept. This is the standard MS MARCO QA / SQuAD evaluation convention.

- **ROUGE-L** — longest-common-subsequence overlap (HF rouge, multi-ref).
- **BLEU** — corpus-level n-gram precision (HF bleu, multi-ref).
- **Exact Match (EM)** — SQuAD-style normalisation, any over references.
- **Token F1** — SQuAD-style token overlap, max over references.

6. Results

Metric	Value
ROUGE-L	0.1619
BLEU	0.0573
Exact Match	0.0050
Token F1	0.1756
# predictions	200

These numbers are not directly comparable to leaderboard MS MARCO QA results because (a) the model is pretrained-only, no SFT, and (b) the conditioning context is the BM25 top-k, not the gold passage.

7. Qualitative Examples

700871 — what is a special educational needs

- **Reference(s)**: 1 behaviour or ability to socialise, eg not being able to make friends. 2 reading and writing, eg they have dyslexia. 3 ability to understand things. 4 concentration levels, eg they have Attention Deficit Hyperactivity Disorder.
- **Prediction**: Special educational needs (SEN) teacher. Special educational needs (SEN) teachers work with children and young people who need extra support

Retrieved context (top passages used):

1. The difference between Special Educational Needs and Disabilities. Children and young people who have special educational needs (SEN) do not necessarily have a disability, and some disabled children and young people do n...
2. Special educational needs teacher. As a special educational needs (SEN) teacher you'll work with children and young people who need extra support, or require an advanced programme of learning in order to reach their full...
3. Special educational needs (SEN) teacher. Special educational needs (SEN) teachers work with children and young people who need extra support. Salary: £24,500 to £70,750 average per year. Hours: 37 per week. 1. Entry requ...

1058325 — what is collier investments inc

- **Reference(s):** Collier Investments Inc locations in Shreveport, LA.
- **Prediction:** Collier Investments, Ltd. was founded in 1999 and is based in Naples, Florida

Retrieved context (top passages used):

1. Find 1 listings related to Collier Investments Inc in Shreveport on YP.com. See reviews, photos, directions, phone numbers and more for Collier Investments Inc locations in Shreveport, LA. Start your search by typing in...
2. Collier Investments, Inc. Dba Manpower has not filed any LCA for H1B visa or LC for green card from fiscal year 2014 to 2016. Collier Investments Dba Manpower had filed 0 LCA and 1 LC from fiscal year 2001 to 2010. Click...
3. Company Overview. Collier Investments, Ltd. is a venture capital firm specializing in early stage investments. The firm typically invests in educational technology companies. It primarily invests in companies based in Un...

1011713 — what was the strawberry kids cereal

- **Reference(s):** Strawberry Shortcake cereal is a crispy, strawberry pink breakfast cereal.
- **Prediction:** Strawberry Shortcake cereal is actually quite delicious and the strawberry taste is fantastic

Retrieved context (top passages used):

1. Strawberry Shortcake Cereal. Strawberry Shortcake cereal is a crispy, strawberry pink breakfast cereal released in 1983 by General Mills. Just like Strawberry Shortcake dolls, Strawberry Shortcake Cereal smells like stra...
2. Huckleberry Pie responds that yes, he has come to help - help himself to a delicious bowl of Strawberry Shortcake cereal! Strawberry Shortcake obliges Huckleberry, who remarks that the cereal is a pretty strawberry pink....
3. cereal box craft for kids, art recycled kids, kids collage ideas, process art kids, scrap art for kids, box art, cereal box crafts for kids, preschool art, recycle activities for kids, recycled art projects for kids, art...

872655 — what language jamaica speak

- **Reference(s):** English and Jamaican.
- **Prediction:** English is the official language of Jamaica, but Jamaican Patois (Patwah) is the spoken language of Jamaica

Retrieved context (top passages used):

1. English is the official language of Jamaica, but Jamaican Patois (Patwah) is the spoken language of Jamaica. Most people in Jamaica speak English and patois, but there are people who only speak Patois.
2. English is the official language of Jamaica, but Jamaican Patois (Patwah) is the spoken language of Jamaica. Most people in Jamaica speak English and patois, but there are people who only speak Patois. The Jamaican langu...

3. English is the official language of Jamaica, but Jamaican Patois (Patwah) is the spoken language of Jamaica. Most people in Jamaica speak English and patois, but there are people who only speak Patois. The Jamaican language...

1096739 — how many pounds of fat can you lose in a week

- **Reference(s):** It's certainly possible to lose more than two pounds per week.
- **Prediction:** How much weight can you lose in a week, How to lose weight fast in 1 week, lose 5 pounds in one week. n no time, you could be hitting your weight loss goal

Retrieved context (top passages used):

1. If you do this for 7 days, you can lose 1 pound of fat in a week. Many experts believe you should not try to lose more than 2 pounds per week. Losing more than 2 pounds in a week usually means that you are losing water w...
2. How much weight can you lose in a week, How to lose weight fast in 1 week, lose 5 pounds in one week. n no time, you could be hitting your weight loss goal. Burn More Calories is a sure way to lose 5 pounds in one week, a...
3. How to lose 10 pounds in a week – the diet plan. Despite the fact that you may not be able to lose 10 pounds in a week, there are many things you can do to start losing weight. It may take more than a week to lose 10 pou...

8. Error Analysis

Predictions with token-F1 = 0 against every reference (a strong signal that the answer is wrong rather than just paraphrased):

1091206 — putrescine definition

- **Reference(s):** Putrescine is a foul-smelling organic chemical compound $\text{NH}_2(\text{CH}_2)_4\text{NH}_2$ that is related to cadaverine.
- **Prediction:** English definition and synonym dictionary from Reverso

1078731 — withdrawal symptoms for coffee

- **Reference(s):** Headaches, irritability, lethargy, nervousness, and mild depression.
- **Prediction:** caffeine withdrawal symptoms

76283 — can you catch a sinus infection

- **Reference(s):** Yes, but not usually.
- **Prediction:** anyone can catch a sinus infection, but certain groups of people are more likely to develop sinusitis

870861 — what kind of sword does Strider Hiryu use

- **Reference(s):** Tonfa-like plasma sword
- **Prediction:** Strider Hiryu is a character from the arcade game Strider

61623 — can Climbing Roses grow as bush roses?

- **Reference(s):** Yes
- **Prediction:** grow between 2 and 6 feet tall

9. Limitations

- The generation model is not fine-tuned on MS MARCO QA. Reference answers are short while pretrained T5/BART tend to produce longer, extractive-style outputs, which depresses overlap-based scores.
- Conditioning on BM25 top-k means retrieval errors propagate directly into generation. A reranking step (week 5) is expected to lift these numbers.
- Surface-form metrics (ROUGE/BLEU/EM) penalise valid paraphrases. Adding a semantic similarity metric (e.g. BERTScore) would give a fairer signal.

10. Next Steps

- Add a cross-encoder reranker between retrieval and generation.
- Add semantic generation metrics (BERTScore, optional NLI-based faithfulness).
- Supervised fine-tuning of the generation model on (question, gold passage, answer) triples drawn from MS MARCO QA v2.1.